AI360

TEAM № 20

Neural Thickets:
Diverse Task Experts Are Dense Around Pretrained Weights

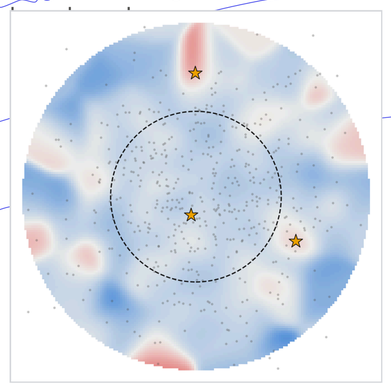
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[1]ITMO · [2]MIPT

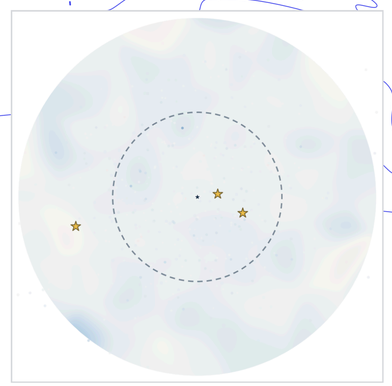
MENTOR Dinar Saberov, GigaChat Pretrain

RANDOPT ON QWEN2.5-3B-INSTRUCT, σ = 0.001: FOUR BENCHMARKS

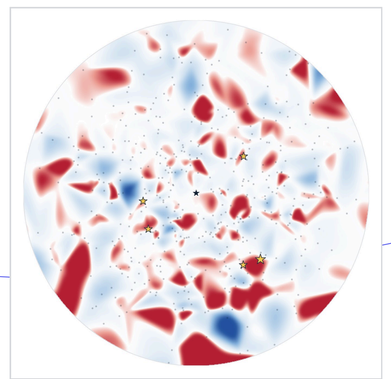
Dot = one weight perturbation · colour = test accuracy change vs base (red above, blue below) · stars = top-3 seeds by train score (small dark star = base model) · own colour scale per



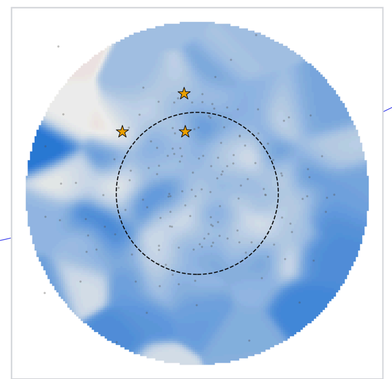
Chemistry · USPTO-50k



Coding · MBPP



Math · GSM8K



Math · MATH-500

How RandOpt works

1. **Perturb.** Sample N random seeds; each defines a model θ + σ·ε with Gaussian noise ε on all weights (σ = 0.001).
2. **Evaluate.** Score every perturbed model on a small train set (200 problems).
3. **Select.** Keep the top-K models by train score (K = 50).
4. **Ensemble.** The K models answer each test problem; the majority vote is the final answer.

No gradients, no training: every seed is independent, so the search is fully parallel.

RandOpt results, Qwen2.5-3B, σ = 0.001

Benchmark	Seeds	Base	Top-1	Top-5	Top-50
Chemistry (USPTO-50k)	500	6.8	13.5	12.8	9.6
Coding (MBPP)	500	57.0	59.0	59.2	62.4
Math (GSM8K)	500	84.2	—	—	—
Math (MATH-500)	159	61.0	61.3	66.3	69.0

Test accuracy, %. Top-K = majority vote of the K best seeds by train score. GSM8K: best single seed (by test) 86.7% at σ = 0.001, 87.0% at σ = 0.0005, 84.5% at σ = 0.002; top-K votes not measured. MATH-500: the 159 of 500 seeds with σ = 0.001.

MATH-500, σ = 0.001: SCALING FROM 3B TO 32B

One colour scale for all models (see bar) · dot = one weight perturbation · stars = top-3 seeds by train score · red = above base, blue = below

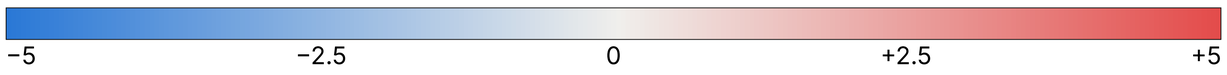
RandOpt on MATH-500 by model size

Model	Seeds	Base	Best seed	Top-1	Top-5	Top-50	Above base
Qwen2.5-3B	300	61.0	62.7	61.3	66.3	69.0	8%
Qwen2.5-7B	300	65.7	72.0	66.7	73.0	76.3	52%
Qwen2.5-14B	300	69.3	70.0	68.3	74.3	78.7	0%
Qwen2.5-32B	300	70.7	75.7	71.0	75.0	78.7	57%

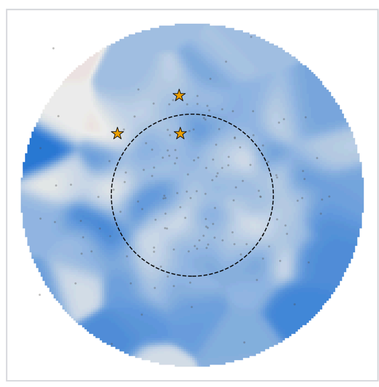
The top-50 ensemble adds +8 to +11 points at every size, although only 0–57% of single perturbations match the base model.

Test accuracy on the 300 MATH-500 test problems, %; only seeds with σ = 0.001. Top-K = majority vote of the K best seeds by train score. 14B heatmap re-coloured from its own ±12 scale to the common scale (all its seeds fall below base).

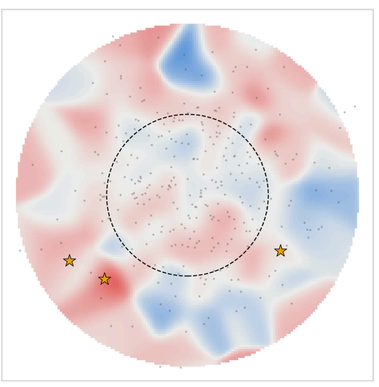
Colour scale (all four models)



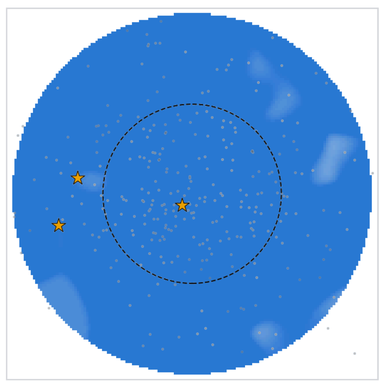
test accuracy change vs the model's own base, percentage points



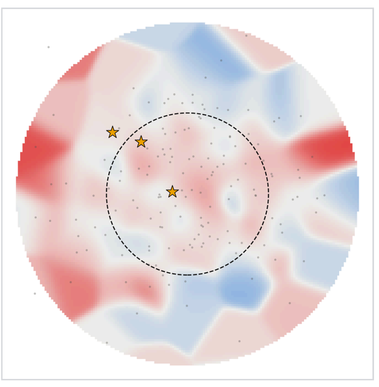
Qwen2.5-3B



Qwen2.5-7B



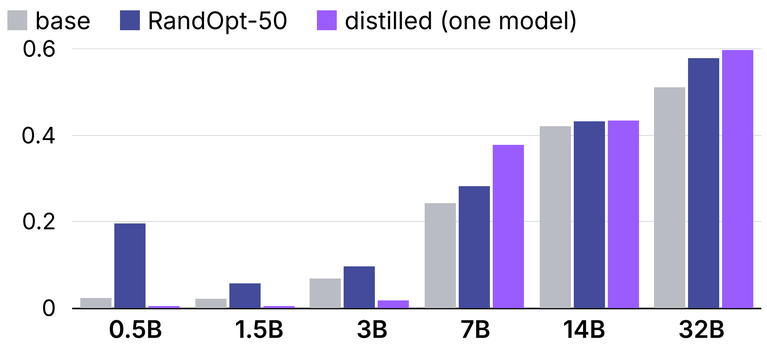
Qwen2.5-14B



Qwen2.5-32B

DISTILLATION OF THE ENSEMBLE AND LORA ATTEMPTS

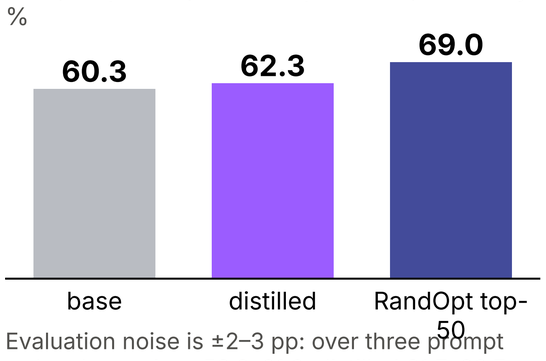
Chemistry (USPTO-50k): test accuracy



model	0.5B	1.5B	3B	7B	14B	32B
base	0.023	0.021	0.068	0.242	0.421	0.510
RandOpt-50	0.195	0.056	0.096	0.281	0.431	0.578
distilled	0.000	0.002	0.016	0.377	0.434	0.597

MATH-500, Qwen2.5-3B

Distilling the ensemble into one model (paper recipe: top-50 seeds, hard problems, LoRA r = 64, 2 epochs); 300 test problems, accuracy %



Evaluation noise is ±2–3 pp: over three prompt orders base gives 59.3–61.3, distilled 61.7–63.7. The gain is within the noise.

LoRA: 32B answers → 3B

test accuracy, %: no LoRA vs after the last epoch

Training set	No LoRA	After	Δ, pp
482 examples, r = 16, 3 epochs			
MATH-500 (200)	63.0	59.5	−3.5
GSM8K (500)	80.4	79.4	−1.0
2475 hard examples (MATH + GSM8K train)			
MATH-500 (200)	60.5	60.5	0.0
GSM8K (500)	83.0	81.2	−1.8
MATH-500, all 500	62.2	60.4	−1.8

For reference, 32B (MATH-500 / GSM8K): base 74.0 / 88.4, top-50 vote 84.0 / 93.6.

No gain: changes are within evaluation noise.